File: main.py

```
#!/usr/bin/env python3

import subprocess
import re
import sys


def banner():
    print("=" * 60)
    print("                    CYBERRECONX")
    print("                NETWORK RECON TOOLKIT")
    print("=" * 60)


def check_tool(tool):
    result = subprocess.run(
        ["which", tool],
        capture_output=True,
        text=True
    )

    return result.returncode == 0


def run_command(command):
    try:
        return subprocess.run(
            command,
            capture_output=True,
            text=True
        )
    except FileNotFoundError:
        return None


def local_network_recon():
    print("\n[*] Fetching local network configuration...")

    if not check_tool("ifconfig"):
        print("[-] Error: ifconfig is not installed.")
        print("[!] Run: sudo apt install net-tools")
        return

    if not check_tool("ipcalc"):
        print("[-] Error: ipcalc is not installed.")
        print("[!] Run: sudo apt install ipcalc")
        return

    result = run_command(["ifconfig"])

    if result is None or result.returncode != 0:
        print("[-] Error: Could not run ifconfig.")
        return

    ip_matches = re.findall(
        r"inet (\d+\.\d+\.\d+\.\d+)",
        result.stdout
    )
```

How main.py runs

```
Interactive Python CLI: shows a banner, offers local or custom network recon, uses system tools (ifconfig/i
```

File: install.sh

```
#!/bin/bash

set -euo pipefail

echo "=============================================="
echo "           CyberReconX Installer"
echo "=============================================="

echo "[*] Updating package list..."
sudo apt update

echo "[*] Installing required tools..."

sudo apt install -y \
    python3 \
    python3-venv \
    python3-pip \
    nmap \
    whois \
    whatweb \
    ipcalc \
    net-tools

echo ""
echo "[+] Installation completed."
echo "[+] Next steps:"
echo " 1) Create a virtualenv: python3 -m venv .venv"
echo " 2) Activate it: source .venv/bin/activate"
echo " 3) Install Python deps (if any): pip install -r requirements.txt"
echo " 4) Run: python3 main.py"
echo ""
echo "Note: Run this script with sudo only when prompted."

exit 0
```

How install.sh runs

```
Bash installer for Debian-based systems: updates apt and installs python3, virtualenv dependencies and netw
```

File: install.ps1

```
# PowerShell installer for Windows
Set-StrictMode -Version Latest

Write-Host '=============================================='
Write-Host '           CyberReconX Installer (Windows)'
Write-Host '=============================================='

Write-Host '[*] Checking for Python...'
$python = Get-Command python -ErrorAction SilentlyContinue
if (-not $python) {
    Write-Host 'Python not found. Please install Python 3.10+ from https://www.python.org/downloads/' -Fore
    exit 1
}

Write-Host '[*] Creating virtual environment (.venv)...'
python -m venv .venv

Write-Host "[*] Activating virtual environment and installing dependencies..."
.
.\.venv\Scripts\Activate.ps1
pip install --upgrade pip
if (Test-Path requirements.txt) {
    pip install -r requirements.txt
}

Write-Host '[+] Installation completed.' -ForegroundColor Green
Write-Host 'Run the tool with: .\.venv\Scripts\Activate.ps1; python main.py'

exit 0
```

How install.ps1 runs

```
PowerShell installer: checks Python availability, creates a virtualenv, activates it, and installs Python d
```

File: requirements.txt

```
# CyberReconX does not require external Python packages.
# Required system tools are installed through `install.sh` (Linux) or
# `install.ps1` (Windows). Leave this file empty if there are no Python
# dependencies, or list them here one per line for `pip install -r`.
```

How requirements.txt runs

```
Lists Python package dependencies for the project (may be empty if none required).
```

File: java/CyberReconX.java

```
// Minimal Java starter for CyberReconX
public class CyberReconX {
    public static void main(String[] args) {
        System.out.println("CyberReconX - Java starter");
        System.out.println("This repository primarily contains Python code.\nKeep Java additions minimal as
    }
}
```

How java/CyberReconX.java runs

```
A minimal Java starter 'CyberReconX' that prints a small banner; added as a placeholder per request and doe
```